# The Shared Discovery Paradox

How a One-Answer Rule Turns Better Information into Worse Search

**Abstract**

Organizations may pool evidence without coordinating how it is used. We study the resulting allocation problem in a Bayesian discovery benchmark with one valuable state and a fixed number of simultaneous actions. Shared reports improve the best single recommendation, yet a rule that repeats that recommendation can reduce discovery by discarding coverage. In a sixteen-state, eight-searcher example, private clue-following discovers the target with probability 0.8322, pooled consensus with 0.3835, and a distinct-action portfolio using the same pooled reports with 0.8594. Seven coordinated actions recover the private benchmark. With a latent common source and nondegenerate signal accuracy, the equal-budget planner gain increases strictly with copying. Equal-split rewards supply partial dispersion at the anonymous symmetric equilibrium, but do not generally recover the planner portfolio; exclusive rewards implement it in pure equilibria under the stated assumptions. We connect these incentives to existing covering and search games rather than claim a new reward mechanism. The contribution is a tractable comparison of information access, assignment authority, incentives and source dependence. The institutional implication is conditional: organizations that share evidence also need an allocation rule, particularly when nominally separate reports reflect common sources. The model does not identify the effect of information sharing alone or validate behavior in actual organizations.

**Keywords:** organizational design, information sharing, search, common-source dependence.
**JEL codes:** C72, D23, D83, O31.

## 1 Introduction

A group can become better informed and less likely to discover what it is looking for. The apparent contradiction arises when an institution solves an estimation problem and then uses the answer as if it had solved an allocation problem. Pooling information may identify the most likely target more accurately. Yet if many available actions are all assigned to that one target, the organization converts a portfolio of attempts into a repeated bet.

This paper isolates that mechanism in a deliberately small environment. Sixteen boxes contain one jackpot. Eight searchers receive noisy clues, each naming the correct box with probability 0.20. If searchers independently follow their own clues, group discovery is 0.8322. If they pool the clues, identify the posterior mode, and all follow it, the selected box is more accurate than any individual clue, but discovery falls to

0.3835. If a coordinator instead assigns the same eight available actions to the eight highest-posterior boxes, discovery rises to 0.8594. Even a coordinator who ignores the clues entirely and opens eight distinct boxes succeeds half the time: no information at all beats informed consensus. The same information supports both failure and improvement; the action protocol determines which one is realized.

The paper develops four layers of the benchmark.

First, we separate the value of information from the quality of the action protocol. For an information set $\mathcal{F}$ and action budget $L$, the attainable frontier is the posterior mass of the best $L$-state portfolio. Protocol loss is the gap between that frontier and the set actually searched. In the canonical pooled-information environment, the entire 47.60 percentage-point gap between consensus and the optimal eight-action portfolio is protocol loss. The resulting action-budget frontier yields an integer recovery budget: seven coordinated actions are sufficient to replace eight decentralized clue-followers.

Second, we ask whether decentralized incentives can supply the missing dispersion. After reports are pooled, $N$ self-interested searchers choose boxes and split a unit prize equally if several select the jackpot. This is a singleton congestion or covering game. It has an exact potential, and its anonymous symmetric equilibrium takes a water-filling form: posterior mass is traded off against expected prize dilution. The equilibrium improves on consensus at every posterior, but it can still collide excessively. Averaging over the canonical report distribution gives discovery 0.5991 and only 2.67 distinct searches in expectation. Thus the canonical ordering becomes

$$G_{\text{consensus}} < G^{\text{sym}}_{\text{market}} < G_{\text{private}} < G_{\text{planner}}.$$

We include the specialized exact bound $2 - 1/N$ for all mixed Nash equilibria. This bound is known in the covering-game literature; the contribution here is a short search-specific proof and its integration into the Bayesian benchmark. Under a uniform posterior on $N$ boxes, anonymous symmetric mixing gives the familiar ratio converging to $e/(e-1)$, even though efficient asymmetric pure equilibria coexist.

Third, we apply the established exclusive-reward correction. Under a sole-rescue rule, the realized prize is paid only if exactly one searcher chose the target. The expected payoff to an agent is then exactly the posterior mass that the agent contributes uniquely to coverage. The resulting potential is social discovery itself. With at least as many boxes as searchers and positive posterior mass on every box, every pure Nash equilibrium assigns one searcher to each of the $N$ highest-posterior boxes. This is full pure-strategy implementation of the planner portfolio, not a dominant-strategy claim: equilibrium selection and mixed behavior remain separate issues.

Fourth, we introduce a latent common-cue copying model. Searchers may repeat the same unobserved source rather than draw independent clues. Copying reduces the number of effective discovery channels, can make raw report counts mis-rank posterior probabilities, and raises the value of reallocating duplicated capacity. We prove that the equal-budget centralized planner gain is strictly increasing in the copying probability. We then run the equal-split game on the herd-aware posterior. In the canonical environment, decentralized report-following initially dominates the symmetric market because independent clues create natural coverage. The ranking reverses at $c = 0.788462$: beyond that point, channel collapse harms private report-following more than posterior concentration harms the dispersing market.

The ingredients have substantial precedents. Team theory, optimal search, parallel R&D, organizational learning, information cascades, epistemic networks, potential games, and covering games each formalize part of the design problem [Marschak and Radner, 1972, Koopman, 1957, Nelson, 1961, March, 1991, Banerjee, 1992, Bikhchandani et al., 1992, Zollman, 2007, Monderer and Shapley, 1996, Gairing, 2009]. We do not claim that communication-induced convergence, correlated signals, congestion-game inefficiency, or marginal-contribution incentives are individually new. The contribution is to place them in one exactly solvable Bayesian discovery benchmark and provide a common set of quantities: one-action value, portfolio value, protocol loss, recovery budget, effective channel count, anonymous-market value, price of anarchy, and rescue implementation.

## 2 The sixteen-box benchmark

There are $M = 16$ boxes, exactly one of which contains a jackpot. The true box is $\theta \in \{1, \ldots, 16\}$ and has a uniform prior. There are $N = 8$ searchers. Conditional on $\theta$, searcher $i$ observes a clue $X_i$ satisfying

$$\mathbb{P}(X_i = \theta \mid \theta) = p = 0.20, \qquad \mathbb{P}(X_i = b \mid \theta) = q = \frac{1-p}{M-1} = \frac{0.8}{15} \quad (b \neq \theta).$$

Throughout the nondegenerate atomic model, $M \geq 2$, $1/M < p < 1$, $q = (1-p)/(M-1) > 0$, and $2 \leq N \leq M$, unless a boundary case is stated explicitly. The clues are conditionally independent in the baseline model. Each searcher can open one box, and the group succeeds if at least one opened box is the target.

### 2.1 Three benchmark protocols

**Private clue-following.** Each searcher opens the box named by the searcher's own clue. The group succeeds whenever at least one clue is correct, so

$$G_{\text{private}} = 1 - (1 - p)^N = 1 - 0.8^8 = 0.83222784. \tag{1}$$

**Shared consensus.** All reports are pooled and every action is assigned to the posterior mode. With a uniform prior and symmetric independent signals, posterior rank equals report-count rank, so consensus is plurality with uniform tie-breaking. Exact enumeration gives

$$G_{\text{consensus}} = 0.38346871. \tag{2}$$

The recommendation is substantially better than one private clue, since $0.3835 > 0.20$. But eight identical actions cover one box, so recommendation accuracy and group discovery coincide.

**Shared portfolio.** A coordinator observes the same pooled reports and assigns actions to the $L$ highest-posterior boxes. With $L = 8$,

$$G_{\text{planner}} = 0.85942125. \tag{3}$$

The planner uses the information gain without compressing the action budget.

These values establish the basic paradox:

$$0.20 < G_{\text{consensus}} = 0.3835 < G_{\text{private}} = 0.8322 < G_{\text{planner}} = 0.8594.$$

Every participant can follow a better-informed recommendation while the group becomes less likely to discover the target.

**Definition 1** (Shared Discovery Paradox)**.** A Shared Discovery Paradox occurs when a shared-information protocol raises average action quality but lowers union success. If protocol $\Pi$ produces actions $A_1, \ldots, A_L$, define average action quality

$$Q(\Pi) = \frac{1}{L} \sum_{\ell=1}^{L} \mathbb{P}(A_\ell = \theta)$$

and discovery $G(\Pi) = \mathbb{P}(\theta \in \{A_1, \ldots, A_L\})$. The paradox is

$$Q(\Pi^S) > Q(\Pi^P) \quad \text{and} \quad G(\Pi^S) < G(\Pi^P).$$

In the canonical comparison, $Q(\Pi^S) = 0.3835 > 0.20 = Q(\Pi^P)$, while $G(\Pi^S) = 0.3835 < 0.8322 = G(\Pi^P)$.

Average action quality admits an exact accounting link to discovery. Let $K = \sum_{\ell=1}^{L} \mathbf{1}\{A_\ell = \theta\}$ be the number of actions that select the target. Then $LQ(\Pi) = \mathbb{E}[K]$ and $G(\Pi) = \mathbb{P}(K \geq 1)$, so

$$LQ(\Pi) - G(\Pi) = \mathbb{E}\big[(K-1)^+\big]. \tag{4}$$

The right side is the expected number of redundant successful actions. Under consensus $K \in \{0, 8\}$, so $8Q - G = 7G = 2.684281$: a correct recommendation produces eight identical hits. Under any collision-free portfolio at most one selected box can be the target, so the gap is zero. Private clue-following lies between the extremes, with $8Q - G = 1.6 - 0.832228 = 0.767772$, because several clues can be simultaneously correct. The paradox is therefore a statement about duplication: a shared-information protocol can raise $Q$ precisely by concentrating $K$.

**Blind-portfolio control.** A blind control calibrates how expensive that duplication is. With a uniform prior and no clues at all, a coordinator who assigns the eight actions to eight distinct boxes succeeds with probability

$$G_{\text{blind}} = \frac{N}{M} = \frac{8}{16} = 0.50.$$

A completely uninformed but diversified portfolio therefore already outperforms informed consensus at 0.383469. This is not a value-of-information comparison—the two protocols observe different things—but it makes the action-compression loss transparent: eight coordinated searches that ignore the clues entirely outperform eight informed searches that all repeat the pooled recommendation.[1] We treat 0.50 as a calibration line rather than a member of the institutional ordering, because blind search discards the available information.

[1]Even eight *uncoordinated* blind guesses, drawn independently and uniformly, succeed with probability $1 - (1 - 1/16)^8 = 0.403281 > 0.383469$.

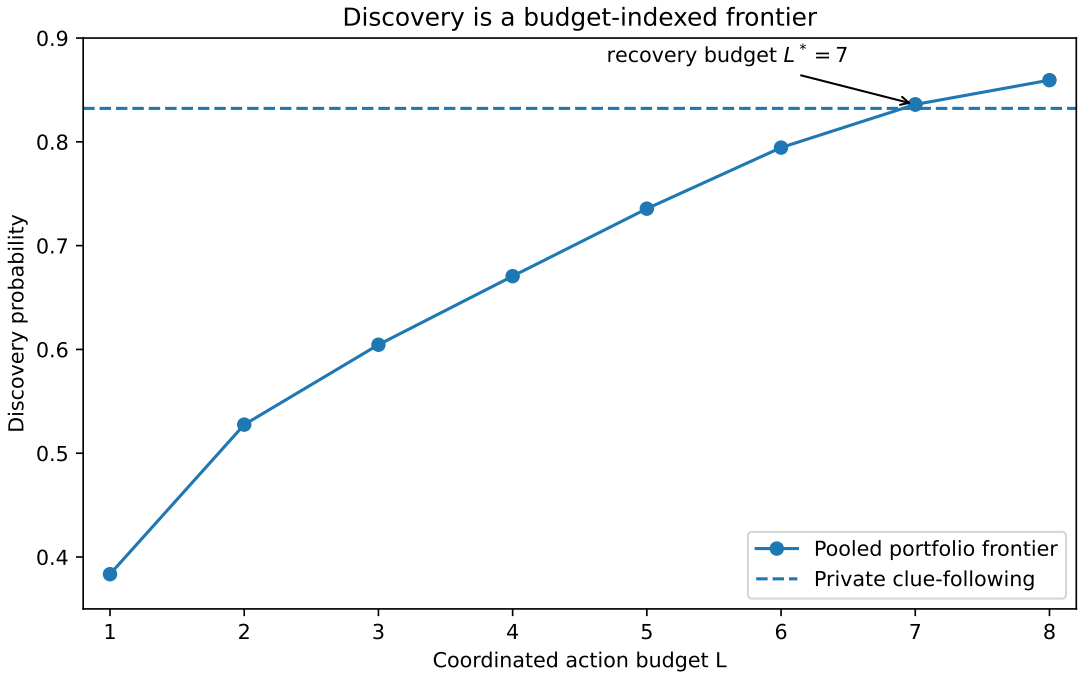



**Figure 1:** The exact pooled-information frontier for the canonical benchmark. The one-action consensus value is only the left endpoint. Seven coordinated actions recover the performance of eight decentralized clue-followers.

## 2.2 The full action-budget frontier

Consensus evaluates pooled information at a one-action budget. Discovery is instead a budget-indexed frontier. Exact enumeration gives

| Actions $L$ | Pooled discovery | Actions $L$ | Pooled discovery |
|---|---|---|---|
| 1 | 0.383469 | 5 | 0.735604 |
| 2 | 0.527532 | 6 | 0.794435 |
| 3 | 0.604528 | 7 | 0.836010 |
| 4 | 0.670581 | 8 | 0.859421 |

The smallest coordinated budget that matches private clue-following is therefore

$$L^* = 7, \qquad G_{\text{portfolio}}(7) = 0.836010 > 0.832228 = G_{\text{private}}.$$

A second measure is the expected number of distinct actions. Private clue-following generates

$$D_{\text{private}} = \left[1 - (1-p)^N\right] + (M-1)\left[1 - (1-q)^N\right] = 6.156850. \tag{5}$$

Consensus generates exactly one distinct action; the planner generates eight. These counts make the mechanism visible: pooling improves the ranking, while the protocol determines how much of the nominal action budget survives as coverage.

## 3 Information value and protocol loss

Let the target be $\theta \in \Omega$ with prior $\mu$. A decision architecture observes an information sigma-field $\mathcal{F}$. A protocol $\Pi$ maps its information into an action profile $(A_1, \ldots, A_L)$, where repeated actions are allowed. Let $S_\Pi = \{A_1, \ldots, A_L\}$ be the set of distinct searched states. Discovery is

$$G_L(\Pi) = \mathbb{P}(\theta \in S_\Pi).$$

For information $\mathcal{F}$ and budget $L$, define the attainable frontier

$$V_L(\mathcal{F}) = \mathbb{E}\left[\max_{S \subseteq \Omega,\ |S| \le L} \mathbb{P}(\theta \in S \mid \mathcal{F})\right]. \tag{6}$$

In an atomic state space, the maximizing set contains the $L$ largest posterior probabilities. Define protocol loss

$$\mathcal{L}_L(\Pi; \mathcal{F}) = V_L(\mathcal{F}) - G_L(\Pi). \tag{7}$$

Thus

$$G_L(\Pi) = V_L(\mathcal{F}) - \mathcal{L}_L(\Pi; \mathcal{F}).$$

This accounting identity separates the quality of available information from the efficiency with which an institution converts information and action capacity into coverage.

**Proposition 1** (Value of information)**.** *If $\mathcal{F}_1 \subseteq \mathcal{F}_2$, then $V_1(\mathcal{F}_2) \ge V_1(\mathcal{F}_1)$.*

*Proof.* Every one-action rule measurable with respect to $\mathcal{F}_1$ remains feasible under $\mathcal{F}_2$. The optimal rule under the richer information cannot perform worse. □

**Proposition 2** (Fixed-budget optimality)**.** *For every realization of $\mathcal{F}$, the posterior top-$L$ portfolio weakly dominates every protocol using at most $L$ distinct actions and the same realized information. Consequently, $V_L(\mathcal{F}) \ge G_L(\Pi)$.*

*Proof.* The set generated by $\Pi$ is one feasible set in the maximization defining $V_L(\mathcal{F})$. The posterior top-$L$ set maximizes conditional target mass. Take expectations. □

In the canonical pooled-information regime,

$$V_8(\mathcal{F}^S) = 0.859421, \qquad \mathcal{L}_8(\Pi_{\text{consensus}}; \mathcal{F}^S) = 0.859421 - 0.383469 = 0.475952.$$

Conditional on the same pooled reports and the same nominal budget, the full gap between consensus and the planner is protocol loss. The comparison with private clue-following changes both the information architecture and the assignment authority, so it is a combined organizational benchmark rather than a unique causal decomposition.

### 3.1 Independent symmetric signals

Let $C_b = \sum_i \mathbf{1}\{X_i = b\}$ be the report count for box $b$. Under the uniform prior,

$$\mathbb{P}(\theta = b \mid C) \propto p^{C_b} q^{N - C_b} = q^N \left(\frac{p}{q}\right)^{C_b}. \tag{8}$$

Because $p > q$, posterior rank equals count rank. Consensus is plurality; the optimal pooled portfolio is the set of boxes with the largest counts, with randomization at cutoff ties.

**Proposition 3** (Private coverage dominates one-action consensus)**.** *In the symmetric independent atomic model,*

$$G_{\text{consensus}} \leq G_{\text{private}} = 1 - (1 - p)^N.$$

*Proof.* The plurality winner has at least one report. Therefore the event that consensus selects the target is a subset of the event that at least one private report names the target. □

The inequality holds throughout the model, but the strict Shared Discovery Paradox additionally requires the pooled one-action value to exceed $p$. With two searchers and uniform tie-breaking, pooled plurality accuracy is exactly $p$, so $N = 2$ is a coverage-compression boundary case rather than a strict paradox.

*Remark* 1 (Partial conformity is monotonically harmful). A natural intermediate institution lets only $a$ of the $N$ searchers defer to the pooled recommendation while the remaining $N - a$ follow their own clues. This dial only goes down. The plurality winner always carries at least one report, so it already belongs to the privately proposed set; an adopter therefore adds nothing to coverage and merely deletes their own

proposal. Formally, for nested adopter sets the searched set satisfies $S_a \subseteq S_{a-1}$ realization by realization, so $G(a)$ is weakly decreasing from $G(0) = G_{\text{private}}$ to $G(N) = G_{\text{consensus}}$; which searchers adopt is immaterial by exchangeability. The decline is strict whenever the displaced clue is uniquely held, correct, and differs from the realized winner, an event of positive probability at every step. Exact enumeration at the canonical parameters gives

$$G(a) = 0.8322,\ 0.7913,\ 0.7461,\ 0.6967,\ 0.6429,\ 0.5848,\ 0.5222,\ 0.4551,\ 0.3835$$
$$(a = 0, 1, \ldots, 8).$$

even the first adopter costs 4.1 percentage points. Dispersion must be informed (private clues), priced (the market of section 4), or assigned (the planner); it cannot be obtained by dosing conformity.

**Beyond the finite example.** Electronic Companion, Section EC.5 states and derives a proportional sparse-evidence limit. It keeps the likelihood ratio fixed as the number of states and searchers grows, rather than holding clue accuracy fixed. The canonical five-protocol ordering persists in that limit. Under fixed nonzero clue accuracy instead, the defined clue-using protocols converge to discovery probability one. The contrast limits the managerial claim: duplicate-action compression matters when evidence remains sparse relative to the search space, not whenever teams communicate.

## 4 Selfish search and the price of coordination

The planner benchmark assumes authority to assign distinct actions. We now retain the common pooled posterior but remove the assignment rule. Fix a posterior

$$\pi = (\pi_1, \ldots, \pi_M), \qquad \sum_b \pi_b = 1.$$

There are $N$ risk-neutral searchers. Each simultaneously chooses one box. If the jackpot is in box $b$ and $k$ searchers chose $b$, each occupant receives $1/k$; all other payoffs are zero. The unit prize is therefore split equally among successful searchers.

### 4.1 Payoffs, potential, and total welfare

Suppose every other searcher uses the same mixed strategy $s = (s_1, \ldots, s_M)$. If a searcher chooses box $b$, the number of other occupants is $K \sim \text{Binomial}(N-1, s_b)$. The reciprocal-share identity is

$$\mathbb{E}\left[\frac{1}{K+1}\right] = \sum_{j=0}^{N-1} \binom{N-1}{j} s_b^j (1-s_b)^{N-1-j} \frac{1}{j+1} = \frac{1-(1-s_b)^N}{N s_b}. \tag{9}$$

Define

$$\phi(s) = \frac{1-(1-s)^N}{Ns}, \qquad \phi(0) = 1. \tag{10}$$

The expected payoff from choosing $b$ is

$$u_b(s) = \pi_b \phi(s_b). \tag{11}$$

The function $\phi$ is strictly decreasing from 1 to $1/N$. Posterior value attracts searchers; congestion dilutes the prize.

For a pure action profile $a$, let $n_b(a)$ be the number of searchers on box $b$, and let $H_k = \sum_{j=1}^{k} 1/j$ with $H_0 = 0$.

**Proposition 4** (Potential and payout identity)**.** *The equal-split game is an exact potential game with potential*

$$\Phi(a) = \sum_{b=1}^{M} \pi_b H_{n_b(a)}. \tag{12}$$

*At every realized action profile, total searcher payoff equals the discovery indicator. Conditional on $\pi$, expected total payoff therefore equals discovery probability.*

*Proof.* If one player moves from a box with occupancy $k$ to a box with occupancy $\ell$, the player's payoff changes by $\pi_c/(\ell+1) - \pi_b/k$. The potential changes by exactly the same amount. For the payout identity, when the target box is occupied, its occupants split a unit prize and total payoff is one; otherwise total payoff is zero. □

Pure Nash equilibria exist by the potential argument [Rosenthal, 1973, Monderer and Shapley, 1996]. They need not be unique or symmetric. The benchmark below focuses on the anonymous symmetric equilibrium selected when ex ante identical searchers use the same mixed rule. This is an explicit equilibrium-selection restriction: identical independent policies do not use role labels or a shared assignment device. Anonymity alone does not establish that actual teams select this equilibrium. The planner and sole-rescue

benchmarks change the admissible assignments or rewards; they are not estimates of a treatment effect of communication.

### 4.2 The water-filling equilibrium

Order posterior masses as $\pi_{(1)} \geq \pi_{(2)} \geq \cdots$. The endpoint $\pi_{(1)}/N$ matters because it is the payoff when everyone chooses the top box.

**Proposition 5** (Anonymous symmetric equilibrium)**.** *The game has a unique anonymous symmetric equilibrium strategy $s$.*

*(i) If $\pi_{(1)}/N \geq \pi_{(2)}$, every searcher chooses the top box and the common payoff is $\lambda = \pi_{(1)}/N$.*

*(ii) Otherwise there is a unique $\lambda \in (\pi_{(1)}/N, \pi_{(1)})$ such that*

$$s_b = \begin{cases} \phi^{-1}(\lambda/\pi_b), & \pi_b > \lambda, \\ 0, & \pi_b \leq \lambda, \end{cases} \qquad \sum_b s_b = 1. \tag{13}$$

*Every box in the support yields expected payoff $\lambda$; every excluded box yields at most $\lambda$.*

*Proof.* In a symmetric equilibrium, used boxes must yield the same payoff. Since $\phi$ is continuous and strictly decreasing, an active box with $\pi_b > \lambda$ has the unique probability in equation (13); a box with $\pi_b \leq \lambda$ cannot attain payoff $\lambda$ even when approached alone. In the non-pure case, the sum of implied probabilities is continuous and strictly decreasing in $\lambda$, exceeds one at $\pi_{(1)}/N$, and equals zero at $\pi_{(1)}$. The pure-corner condition is exactly the no-deviation condition from the top box to the second-best box. □

The rule is water-filling on the posterior. Searchers flood boxes above a common value threshold, but they do not impose a top-$N$ count cutoff and they still collide. This is the invisible hand's partial substitute for portfolio assignment.

**Proposition 6** (Welfare identity)**.** *At the anonymous symmetric equilibrium,*

$$G^{\text{sym}}_{\text{market}} = \sum_b \pi_b \left[1 - (1 - s_b)^N\right] = N\lambda. \tag{14}$$

*Moreover, $G^{\text{sym}}_{\text{market}} \geq \max_b \pi_b$, with strict inequality unless the equilibrium is pure.*

*Proof.* For an active box, $\pi_b[1-(1-s_b)^N] = Ns_b\pi_b\phi(s_b) = Ns_b\lambda$. Summing and using $\sum_b s_b = 1$ gives $N\lambda$. If $b^*$ is a posterior mode, then

$$N\lambda = \pi_{b^*}\frac{1-(1-s_{b^*})^N}{s_{b^*}} \geq \pi_{b^*},$$

because $1-(1-s)^N \geq s$. Equality requires $s_{b^*} = 1$. □

Thus equal sharing fixes the most extreme consensus pathology posterior by posterior: congestion makes repetition costly. But the game can still under-cover the posterior tail.

### 4.3 The exact price of anarchy

The planner assigns one searcher to each of the $N$ highest-posterior boxes, giving

$$G_{\text{opt}} = \sum_{b=1}^{\min\{N,M\}} \pi_{(b)}.$$

Following Koutsoupias and Papadimitriou [1999], define PoA $= G_{\text{opt}}/G_{\text{eq}}$ for the worst equilibrium. The following sharp bound specializes the uniform-sharing covering-game analysis of Gairing [2009], Theorem 2 (long version); that paper uses the reciprocal welfare ratio. A direct search-specific proof is included for transparency, not claimed as a new general inefficiency theorem.

**Theorem 1** (Exact mixed price of anarchy)**.** *For every posterior $\pi$ and every $N$-searcher equal-split game,*

$$\text{PoA} \leq 2 - \frac{1}{N}. \tag{15}$$

*The bound applies to mixed Nash equilibria and is tight.*

*Proof.* Let $O = \{o_1, \ldots, o_N\}$ be an optimal set, assigning one optimal box $o_i$ to each searcher $i$ (ignore zero-mass padding if $M < N$). For any realized action profile $a$, write $W(a)$ for covered posterior mass. For each $o_i$,

$$\pi_{o_i} \leq \left(1-\frac{1}{N}\right)\pi_{o_i}\mathbf{1}\{o_i \text{ is occupied in } a\} + u_i(o_i, a_{-i}). \tag{16}$$

If $o_i$ is empty, deviating to it gives $\pi_{o_i}$. If it is occupied, deviating gives at least $\pi_{o_i}/N$, which completes the

right-hand side. Summing equation (16) over $i$ gives

$$G_{\text{opt}} \le \left(1 - \frac{1}{N}\right) W(a) + \sum_i u_i(o_i, a_{-i}),$$

because the occupied optimal boxes contribute at most $W(a)$. At a mixed Nash equilibrium, each player's expected current payoff is at least the payoff from the fixed deviation to $o_i$. Total expected current payoff equals $\mathbb{E}[W(a)]$. Hence

$$G_{\text{opt}} \le \left(2 - \frac{1}{N}\right) \mathbb{E}[W(a)].$$

For tightness, let one "decoy" box have posterior mass $1/2$ and let $N$ tail boxes each have mass $1/(2N)$. The profile in which all searchers choose the decoy is a Nash equilibrium: each receives $1/(2N)$ and is indifferent to deviating to an empty tail box. Equilibrium discovery is $1/2$, while the planner covers the decoy and $N - 1$ tail boxes, giving $1 - 1/(2N)$. The ratio is $2 - 1/N$. □

The theorem distinguishes two questions. A posterior may possess an efficient pure equilibrium and an inefficient anonymous mixed equilibrium at the same time. The price of anarchy controls the worst strategic outcome; the symmetric benchmark measures what happens under anonymous, uncoordinated mixing.

The proof is a smoothness argument in the sense of Roughgarden [2015]: inequality equation (16) establishes that the game is $(1,\ 1 - 1/N)$-smooth, so the extension theorems for robust price-of-anarchy bounds apply verbatim. The bound $2 - 1/N$ therefore holds not only for mixed Nash equilibria but for all coarse correlated equilibria and asymptotically for the average play of no-regret learning dynamics—agents need not compute an equilibrium for the guarantee to bind.

**Uniform posterior.** If posterior mass is uniform on $T$ boxes, the anonymous symmetric equilibrium is $s_b = 1/T$ and

$$G^{\text{sym}}_{\text{market}} = 1 - \left(1 - \frac{1}{T}\right)^N, \qquad G_{\text{opt}} = \frac{\min\{N, T\}}{T}. \tag{17}$$

When $T = N$,

$$\frac{G_{\text{opt}}}{G^{\text{sym}}_{\text{market}}} = \frac{1}{1 - (1 - 1/N)^N} \longrightarrow \frac{e}{e - 1} \approx 1.582. \tag{18}$$

For $N = 8$, the ratio is 1.523482. An efficient pure equilibrium placing one searcher on each box also exists. The $e/(e-1)$ constant is therefore the collision cost of symmetric mixing, not a universal impossibility result.

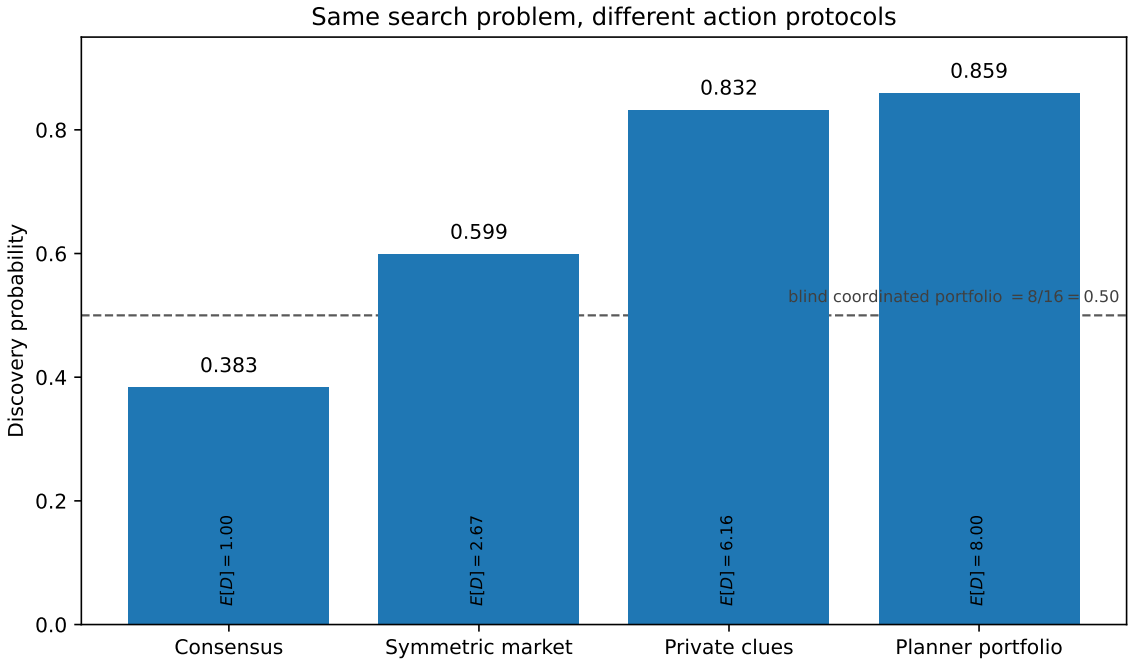



**Figure 2:** Exact ex-ante discovery under four protocols in the independent sixteen-box benchmark. The labels inside the bars report the expected number of distinct boxes searched. The dashed line marks the blind coordinated portfolio at $8/16 = 0.50$: informed consensus falls below a protocol that ignores the clues entirely. Equal sharing supplies partial dispersion, but not enough to match natural coverage from independent private clues.

### 4.4 The canonical four-protocol ordering

For every realized pooled posterior in the sixteen-box benchmark, we solve equation (13) and then average over the exact distribution of report-count classes. The resulting anonymous market discovery is

$$G_{\text{market}}^{\text{sym}} = 0.59909925. \tag{19}$$

It generates 2.673494 distinct actions in expectation. The complete canonical comparison is shown in figure 2 and table 1.

**Table 1:** Canonical protocol comparison: $M = 16$, $N = 8$, $p = 0.20$.

| Protocol | Information | Allocation rule | Quality $Q$ | Discovery $G$ | $\mathbb{E}$[distinct] |
|---|---|---|---|---|---|
| Consensus | pooled | repeat posterior mode | 0.383469 | 0.383469 | 1.000 |
| Symmetric market | pooled | equal-split water-filling | 0.344136 | 0.599099 | 2.673 |
| Private clues | local | follow own report | 0.200000 | 0.832228 | 6.157 |
| Planner portfolio | pooled | top eight, one each | 0.107428 | 0.859421 | 8.000 |

$Q$ is the average standalone probability that a selected box contains the jackpot, as in equation (4). The symmetric-market row is an equilibrium-selection benchmark, not a claim that every Nash equilibrium has the same value.

The table records an exact inversion:

$$Q_{\text{consensus}} > Q^{\text{sym}}_{\text{market}} > Q_{\text{private}} > Q_{\text{planner}} \qquad \text{while} \qquad G_{\text{consensus}} < G^{\text{sym}}_{\text{market}} < G_{\text{private}} < G_{\text{planner}}.$$

Moving toward the group-optimal allocation requires assigning actions that are individually less likely to be correct in isolation: the planner raises group discovery to 0.8594 while reducing the average standalone probability that an assigned box contains the jackpot to 0.1074. Average quality should not be confused with equilibrium payoff. By equation (14), the symmetric market payoff per searcher is $\lambda = G^{\text{sym}}_{\text{market}}/N = 0.074887$. The quality number says the market concentrates on individually attractive boxes; the payoff number says crowding sharply dilutes the private return from doing so.

Congestion pricing cures unanimity but not concentration. Private clue-followers condition on different signals and therefore obtain substantial natural coverage. Symmetric market participants condition on the same posterior and rationally overweight its peak. The invisible hand disperses, but toward the wrong distribution.

## 5 Reward the rescue, not the win

The exclusive-reward policy is studied directly by Collet and Korman [2018], Korman and Rodeh [2020]. We call it sole rescue to emphasize marginal coverage; we do not propose it as a new mechanism. The equal-split rule rewards an agent for being on the jackpot box. The group values an agent for being on a jackpot box that no one else covered. The difference is marginal contribution.

Consider the *sole-rescue rule*: if exactly one searcher chose the realized target, that searcher receives the unit prize; if two or more chose the target, no one is paid. Conditional on posterior $\pi$, a searcher choosing box $b$ receives expected payoff $\pi_b$ if alone on $b$ and zero otherwise.

**Proposition 7** (Pure-strategy implementation)**.** *Suppose $M \geq N$ and every posterior mass is positive. Under*

*the sole-rescue rule:*

*(i) the game is an exact potential game with potential equal to covered posterior mass;*

*(ii) every pure Nash equilibrium is collision-free; and*

*(iii) the occupied boxes in every pure Nash equilibrium are exactly an $N$-box posterior-optimal portfolio.*

*Thus every pure Nash equilibrium attains $G_{\text{opt}}$.*

*Proof.* If a player leaves a box, covered mass falls by $\pi_b$ exactly when the player was its sole occupant. If the player enters another box, covered mass rises by $\pi_c$ exactly when that box was empty. The player's payoff change therefore equals the change in covered mass.

If a pure profile contains a collision, at least one box is empty because $M \geq N$. A colliding player earns zero and can move to an empty positive-mass box for a strictly positive payoff, so no collision can persist. In a collision-free profile, a player on box $b$ earns $\pi_b$. Moving to an occupied box yields zero; moving to an empty box $c$ yields $\pi_c$. The profile is a Nash equilibrium if and only if every occupied box has posterior mass at least as large as every empty box. Its occupied set is therefore a top-$N$ set. □

The result is full implementation in pure Nash equilibrium, not dominant-strategy implementation. A searcher's best action still depends on which boxes others occupy. Nor does the result eliminate all mixed-equilibrium or equilibrium-selection concerns. A public role-assignment or sequential convention may still be useful for selecting among efficient top-$N$ matchings.

The rule is weakly budget-balanced ex post: it never pays more than the unit prize. On the efficient equilibrium path it pays the full prize exactly when discovery occurs. Off equilibrium it burns the prize on a duplicated discovery. In richer applications, attribution, side payments, risk aversion, and collusion may make literal sole-rescue contracting difficult. The benchmark point is narrower: compensating unique coverage aligns the potential with the organization's discovery objective, whereas equal sharing aligns total payout with discovery but not the allocation of effort.

## 6 Common-source dependence

Real searchers rarely receive conditionally independent signals. They read the same reports, use the same models, attend the same meetings, and inherit the same narratives. We introduce one latent common source to make this dependence explicit.

### 6.1 Copying and channel collapse

Let $X_0$ be a common cue with the same accuracy law as a private clue. Each searcher independently copies $X_0$ with probability $c$ and otherwise draws an independent clue:

$$X_i = \begin{cases} X_0, & \text{with probability } c, \\ \widetilde{X}_i, & \text{with probability } 1 - c. \end{cases}$$

The copying probability $c$ is a scalar parameter—distinct from the report-count vector $C = (C_1, \ldots, C_M)$ of section 3.1—and it is not the pairwise correlation coefficient. For correctness indicators $Y_i = \mathbf{1}\{X_i = \theta\}$,

$$\operatorname{Corr}(Y_i, Y_j \mid \theta) = c^2.$$

Let $K \sim \text{Binomial}(N, c)$ be the number of copiers. Conditional on $K$, the copiers contribute one common discovery channel if $K \geq 1$, while the $N - K$ noncopiers contribute independent channels.

**Proposition 8** (Channel collapse)**.** *The effective channel count is*

$$N_{\text{eff}} = N - K + \mathbf{1}\{K \geq 1\}. \tag{20}$$

*Private report-following discovery is*

$$G_{\text{private}}(c) = 1 - p\big[(1-c)(1-p)\big]^N - (1-p)\big[1 - p(1-c)\big]^N. \tag{21}$$

*Proof.* If at least one searcher copies, private search misses only if the common cue misses and every noncopier's independent clue misses. If no searcher copies, all $N$ independent clues must miss. Averaging over $K$ and collecting terms yields equation (21). □

At $c = 0$, equation (21) is $1-(1-p)^N$. At $c = 1$, it is $p$: eight nominal searchers have become one effective clue. In the canonical model, expected distinct proposals fall from 6.157 at $c = 0$ to 1.362 at $c = 0.95$.

### 6.2 Agreement need not mean posterior rank

Under independence, report count is a sufficient ranking statistic. Under copying, a large count can be generated by one incorrect source repeated many times. Exact marginalization over the latent cue and copier count can reverse the ranking.

At $c = 0.4$, consider counts $(6, 2, 0, \ldots, 0)$. Let $a$ be the six-count box and $b$ the two-count box. The exact count likelihoods are

$$\mathbb{P}(C \mid \theta = a) = 1.1427 \times 10^{-4}, \qquad \mathbb{P}(C \mid \theta = b) = 1.3977 \times 10^{-4}.$$

With a uniform prior, the two-count box has the larger posterior. This does not claim that count ranking generally fails dramatically; it establishes that dependence-aware ranking is conceptually distinct from raw agreement. In the canonical calculations, the Bayes-versus-count discovery discount is concentrated at one- and two-action budgets and vanishes for $L \geq 3$ on the tested grid.

### 6.3 The centralized planner gain

Let $D$ be the number of distinct privately proposed boxes and let $V_N(c)$ be the Bayes-optimal $N$-action portfolio value after observing reports generated at copying probability $c$. Define the equal-budget planner gain

$$\Delta_{\text{planner}}(c) = V_N(c) - G_{\text{private}}(c). \tag{22}$$

**Theorem 2** (Monotone planner gain)**.** *In the atomic $M$-state model with $2 \leq N \leq M$ and $1/M < p < 1$, the planner gain $\Delta_{\text{planner}}(c)$ is strictly increasing for $c \in (0, 1)$. At full copying,*

$$\Delta_{\text{planner}}(1) = (N - 1)q. \tag{23}$$

The proof is in Electronic Companion, Section EC.3. Its key identity is

$$\Delta_{\text{planner}}(c) = \mathbb{E}\left[\mathbf{1}\{\text{private search misses}\}\frac{N - D}{M - D}\right]. \tag{24}$$

The integrand is defined to be zero when $D = M$; that event entails $N = M$ and private success. At $p = 1$, excluded from the strict theorem, the gain is identically zero. All named boxes precede unnamed boxes in the posterior ranking, so the planner preserves every distinct proposal and fills unused slots with otherwise unsearched states. Conditional on a miss, unnamed states are symmetric. Increasing the number of copiers weakly shrinks $D$ and can convert a success into a miss; both changes raise the value of the planner's fill slots. The binomial copier count shifts upward with $c$.

For the canonical parameters, the gain rises from 0.027193 at $c = 0$ to 0.335927 at $c = 0.95$, with limit 0.373333. The recovery budget falls from seven coordinated actions at $c = 0$ to two at $c = 0.95$. The

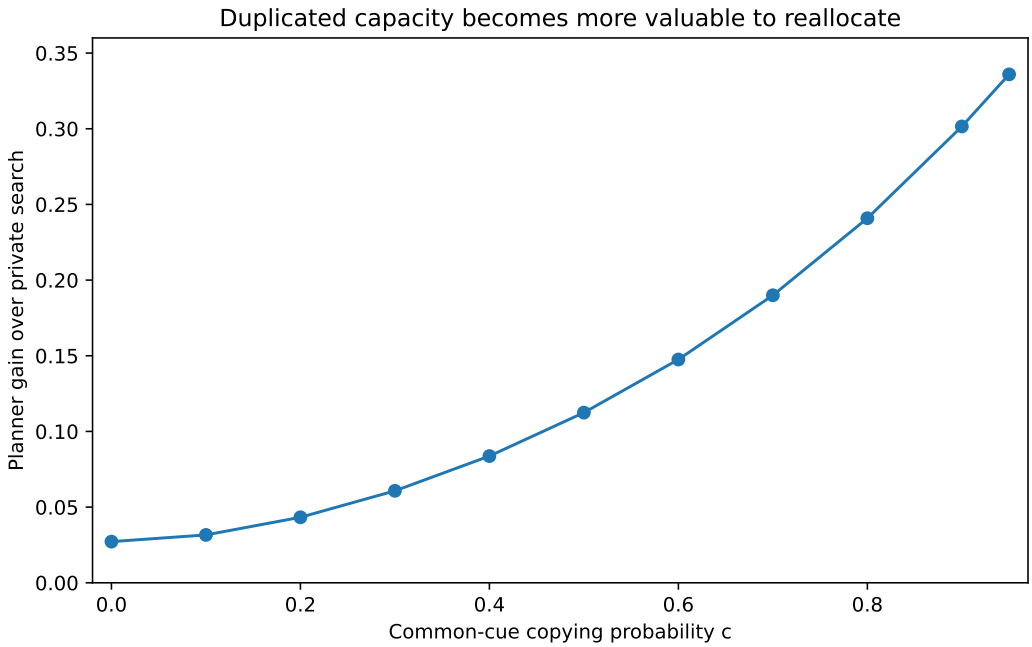



**Figure 3:** Exact canonical planner gain $V_8(c) - G_{\text{private}}(c)$. As common-source copying creates more duplicated capacity, identifying and reallocating it becomes more valuable.

**Table 2:** Canonical outcomes under common-cue copying.

| $c$ | Consensus | Symmetric market | Private reports | Planner |
|---|---|---|---|---|
| 0.00 | 0.383 | 0.599 | 0.832 | 0.859 |
| 0.50 | 0.271 | 0.552 | 0.655 | 0.768 |
| 0.70 | 0.240 | 0.477 | 0.512 | 0.702 |
| 0.80 | 0.221 | 0.428 | 0.423 | 0.664 |
| 0.90 | 0.206 | 0.388 | 0.319 | 0.621 |
| 0.95 | 0.202 | 0.380 | 0.262 | 0.598 |

theorem concerns a frictionless centralized benchmark: it combines pooled observation with assignment authority and therefore upper-bounds what a decentralized institution may realize.

### 6.4 Selfish search under copying

Now let searchers pool their reports, form the herd-aware posterior by marginalizing over the latent cue and copier count, and play the equal-split game on that posterior. For each report-count class, we solve the anonymous equilibrium and average conditional discovery over its exact probability.

At low copying, private report-following wins because different reports supply natural coverage. As copying rises, that advantage collapses. The market continues to disperse over the support of the common

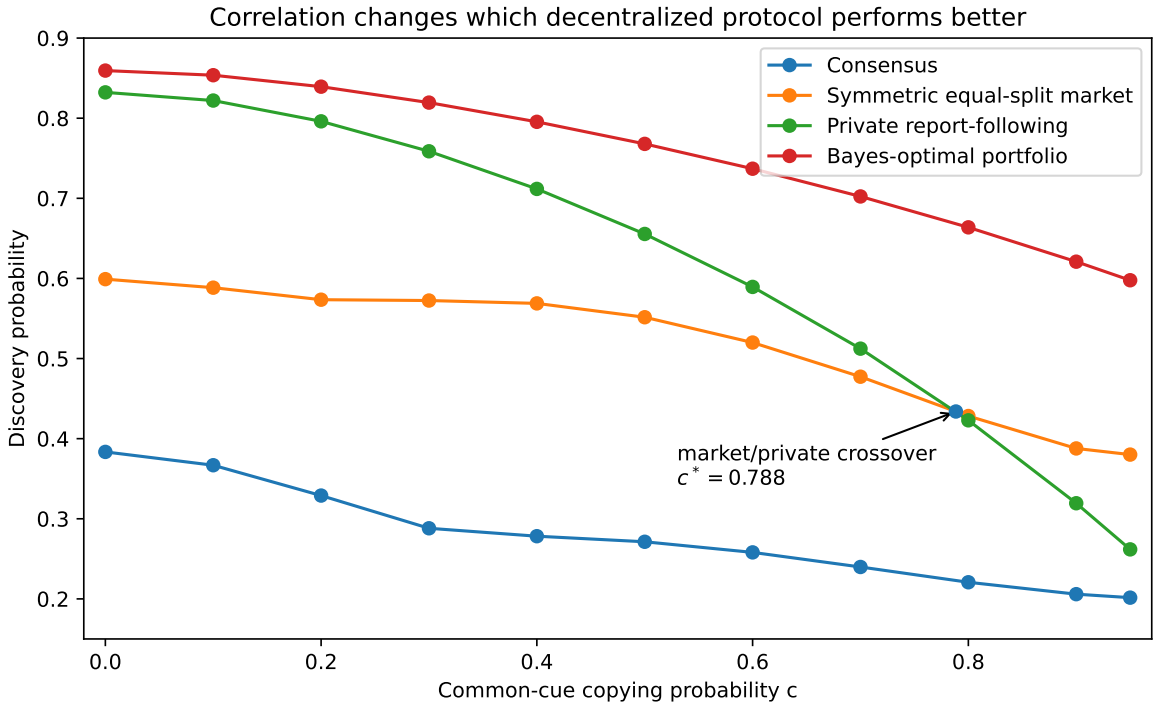



**Figure 4:** Exact canonical comparison under common-source copying. The symmetric equal-split market initially trails decentralized report-following, then overtakes it when channel collapse becomes sufficiently severe. The crossover is a canonical computational result, not a general theorem.

posterior and therefore degrades more slowly. High-precision enumeration gives the canonical crossover

$$c^* = 0.7884616565, \qquad G^{\text{sym}}_{\text{market}}(c^*) = G_{\text{private}}(c^*) = 0.4338934643. \tag{25}$$

At full copying, private report-following and consensus both equal $p = 0.20$, the planner reaches $p + (N-1)q = 0.573333$, and the anonymous equal-split market reaches approximately 0.382006.

The crossover is not knife-edge in the parameters. A comparative-statics scan over 72 combinations—$M \in \{8, 12, 16, 24, 32\}$, $N \in \{4, 6, 8\}$, $p \in \{0.10, \ldots, 0.30\}$ with $p > 1/M$, and $c$ scanned in steps of 0.02 up to 0.98—finds that the market–private gap is negative at $c = 0$ in every combination and changes sign at most once: 52 combinations cross exactly once, at crossovers ranging from $c \approx 0.29$ to $c \approx 0.98$, while in the remaining 20, concentrated at large $M$, small $N$, and strong signals, private report-following stays ahead throughout the scanned range. On this grid the qualitative pattern is uniform—natural coverage first, forced dispersion after enough channel collapse—though single crossing remains a computational pattern rather than a theorem.

The institutional implication is conditional. When information sources are genuinely independent, decen-

tralized local action can dominate a common-posterior market. When sources are highly correlated, some mechanism that forces dispersion can outperform nominal decentralization. Correlation raises the value not only of a perfect planner but also of imperfect coordination.

# 7 Related work

**Team theory and optimal search.** The common-objective model is a team problem in the sense of Marschak and Radner [1972]: agents possess dispersed information and choose a joint action profile. Blackwell's comparison of experiments underlies the value-of-information result [Blackwell, 1953]. The operational parent is optimal search: Koopman [1957] studies how finite search effort should be allocated over uncertain target locations. The atomic model is a discrete, all-or-nothing corner in which the posterior top-$L$ rule is exact.

**Parallel R&D and organizational learning.** Parallel projects hedge uncertainty in innovation [Nelson, 1961]. Related accounts examine organizational learning and the division of cognitive labor [March, 1991, Kitcher, 1990]. Network epistemology models the consequences of communication structure [Zollman, 2007]; Mayo-Wilson et al. [2011] distinguish individual from group epistemic rationality. In the present accounting, limited communication is a second-best instrument: sparse networks preserve diversity only by withholding information value, whereas the planner and rescue benchmarks keep the shared posterior intact and repair the action allocation directly, separating the two effects that communication limits bundle. Duede and Evans [2024] independently describe the belief-side half of this trade-off—collaboration inflates collective certainty while the likelihood of genuine replication falls—a related belief-side tension rather than an equivalence to our action-coverage mechanism. Rzhetsky et al. [2015] study a much richer dynamic problem of experiment choice on an evolving scientific network. The present benchmark strips away dynamics and careers to expose a finite-action accounting identity: a community can improve its shared ranking and still waste its experimental portfolio.

**Herding and correlated judgment.** Information-cascade models explain how rational observation can generate convergent beliefs or actions [Banerjee, 1992, Bikhchandani et al., 1992, Smith and Sørensen, 2000]. Correlated-jury work analyzes dependence in voting aggregation [Ladha, 1992]. Our copying model does not claim the first failure of raw agreement. It places dependence inside a multi-alternative discovery problem and measures three distinct consequences: channel collapse, posterior-versus-count ranking loss, and the

value of reallocating duplicated actions.

**Congestion, covering, and potential games.** The equal-split extension is a singleton congestion game with a harmonic potential [Rosenthal, 1973, Monderer and Shapley, 1996]. Its welfare is a coverage function, linking it to valid-utility and covering games [Vetta, 2002, Gairing, 2009]. This lineage is important for claim calibration: the exact $2 - 1/N$ bound is not presented as a new general result in algorithmic game theory. The benchmark combines these established ingredients by studying the way the covering game is generated endogenously by a Bayesian posterior and compared ex ante with private signals, consensus, a planner, and common-source copying. A complementary line studies how information itself interacts with congestion: Acemoglu et al. [2018] show that giving drivers additional route information can raise equilibrium travel times—an informational Braess' paradox—and characterize the network class immune to it, with a subsequent information-design literature choosing what to disclose [Das et al., 2017]. The mechanism is instructively different. There, information expands strategy sets in a delay-minimization routing game and harms through re-routing externalities in equilibrium; here, the information architecture is held fixed, richer information weakly raises the attainable frontier (the argument of proposition 1 applies at every budget), and the loss arises entirely from the protocol that converts the posterior into actions. Whether a sharper shared posterior can lower equilibrium coverage through the water-filling pure corner—an informational Braess' paradox in discovery rather than delay—is a disclosure-stage question we leave open.

**Direct search-game precedents.** Conditional on the pooled posterior, our stage game is the one-round search game with site values $f(b) = \pi_b$ and $k = N$ players. Collet and Korman [2018] study this weighted coverage setting, establish the unique symmetric equilibrium through ideal-free-distribution conditions, and show that exclusive rewards maximize coverage among symmetric independent profiles. Korman and Rodeh [2020] extend the model to multiple rounds, including the sharing and exclusive policies; our stage game is their $T = 1$ case. The water-filling condition here is a specialized expression of payoff equalization, not a new equilibrium principle. Their results make it essential to distinguish optimal symmetric coverage from the unrestricted top-$N$ assignment. For two equally likely states and two players, independent symmetric mixing covers with probability $3/4$, whereas distinct role assignments cover with probability one. Our short pure-implementation argument does not supersede their symmetric result or establish that every mixed equilibrium is efficient. Kleinberg and Oren [2011] also study project selection and credit design, with project success technologies; unlike our atomic target model, their projects can yield multiple successes. The remaining contribution sought here is the ex ante comparison across signal and assignment protocols, including a

dependence-specific planner-gain characterization, not the novelty of exclusive rewards or coverage games.

**Related follow-on work.** Nakajima [2026] studies an incremental-sharing frontier and a two-agent Bayesian game in a related finite discovery setting. Its aggregation-versus-independent-rescue condition and selected-equilibrium positive-sharing interval are not results claimed by this paper. The present manuscript retains the original fixed-protocol benchmark, copying-dependent planner gain and equal-split/exclusive-reward comparison. The two papers share the information-versus-action-compression motivation, so the latter is not presented here as an exclusive new research theme.

**Submodular extensions.** With one atomic target, the value of a set of distinct boxes is modular. If actions cover overlapping state sets, the objective becomes monotone submodular; if actions reveal information sequentially, adaptive submodularity becomes relevant [Nemhauser et al., 1978, Golovin and Krause, 2011]. The exact top-$L$ portfolio would then be replaced by approximation guarantees, but protocol loss and recovery budgets would remain well-defined.

# 8 Organizational design implications

The decision is how a team with a fixed action budget translates shared reports into assignments. Information access, assignment authority, a mandate to repeat one action, and common-source dependence are separate design choices. Under the planner's authority and the same pooled posterior as consensus, replacing duplicate actions by the top-$N$ portfolio raises canonical discovery by 47.60 percentage points. This comparison identifies protocol loss within the model; it does not compare unrestricted optimal private-information teams with optimal shared-information teams.

The comparative static sharpens that distinction. In the common-cue model, the planner preserves all distinct private proposals and fills the slots released by duplication. As copying rises, the expected gain over private report-following increases strictly under the theorem's nondegenerate assumptions. Nominal decentralization therefore need not preserve diverse actions when workers draw on a common source. Source provenance is relevant to the value of assignment capacity, even though more common sourcing need not improve absolute planner discovery.

The advice has a limit and a reversal. With independent clues the symmetric market discovers less than private clue-following in the canonical example; at high copying their ranking reverses. In the fixed-accuracy abundant-evidence limit, even consensus succeeds with probability tending to one, leaving little level gain to

diversification. Thus this model does not prescribe diversification regardless of evidence or cost. It omits the cost of operating an assignment system, strategic reporting, heterogeneous projects and endogenous adoption of a one-answer rule. A manager cannot infer an optimal institutional investment without those additional quantities.

A research team using one shared ranking could distinguish the quality of its leading hypothesis from the number of distinct tests performed, and compare repeated tests with the posterior mass of available untested alternatives. This is a model-based diagnostic, not a validated performance estimator for research organizations. Venture portfolios involve multiple successes, prices and heavy-tailed returns, and are still farther from the one-target model. Teams using the same AI model or retrieval source are a motivating analogy only: our copying process has not been measured in those systems, and the model's payoff and no-regret assumptions cannot be presumed to hold for them.

# 9 Limitations and extensions

The benchmark is intentionally narrow. Exactly one atomic state is valuable; each action tests one state; actions are simultaneous and costless; private clues are ex ante symmetric; the planner can observe and assign reports; equal-split players share a common posterior; and dependence is generated by one latent cue whose provenance is hidden. These assumptions make both inference and assignment unusually clean.

Several qualifications matter for interpretation. First, the canonical market value uses the anonymous symmetric equilibrium. Equal-split games can also have efficient asymmetric pure equilibria, so role labels, conventions, or sequential play may materially change outcomes. Second, sole-rescue gives full pure-equilibrium implementation but not dominant strategies or a guarantee for every mixed equilibrium. Third, the monotone planner-gain theorem is exact for the stated copying process; the market/private crossover is a canonical computational result. Fourth, the planner benchmark changes both information access and decision rights relative to decentralized clue-following.

One intermediate cell deserves explicit mention. A decentralized-team benchmark would allow the agents to commit ex ante to role-dependent policies $f_1, \ldots, f_N$ while retaining private, noncommunicated clues, with team value $\max_{f_1,\ldots,f_N} \mathbb{P}\big(\theta \in \{f_1(X_1), \ldots, f_N(X_N)\}\big)$. Characterizing that private-information team optimum would separate assignment authority from information pooling, which the planner benchmark changes jointly. We leave it open rather than insert a hand-selected policy: the simple hybrid that replaces clue-followers with blind sentinels is maximized at zero sentinels in the canonical environment, so the inter-

esting policies are signal-contingent.

Natural extensions include heterogeneous searchers, multiple common sources or cliques, endogenous information acquisition, multiple targets, costly or capacity-constrained actions, heavy-tailed prizes, sequential experimentation, overlapping coverage, collusion under rescue rewards, and mechanism design when the posterior itself is privately held. In these environments the numerical yardsticks may change, but the benchmark's separation between information, allocation, incentives, and dependence remains available.

## 10 Conclusion

Collective discovery is not collective estimation with more participants. Estimation asks for one answer. Discovery allocates a portfolio of attempts.

The sixteen-box benchmark makes the distinction exact. Pooling eight imperfect clues raises the best single recommendation from 20.0% to 38.35%. Repeating that recommendation lowers discovery from 83.22% to 38.35%. A planner using the same pooled reports reaches 85.94%, and seven coordinated actions recover the private benchmark. Equal sharing supplies partial dispersion and raises discovery to 59.91%, but its worst equilibrium can lose a factor $2 - 1/N$. Rewarding unique rescue aligns every pure equilibrium with the first-best portfolio. Common-source copying collapses natural coverage, increases the value of reallocation, and eventually makes even an imperfect dispersing market outperform private report-following.

The central design lesson is therefore not "share less information." It is: treat information sharing and action allocation as distinct institutional choices. Share the evidence. Diversify the actions.

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